

CS471 – Lesson 15 Solutions Key

Reinforcement Learning: Model-Free Learning and Q-Learning

Part 1: Guided In-Class Worksheet (Instructor-Led)

1. Model-Based Planning vs. Model-Free Learning

- a. **Question:** Contrast Model-Based Planning (such as Value Iteration) with Model-Free Learning (Q-learning). Identify what environmental information is missing in model-free scenarios.

Solution

Model-Based Planning: Assumes complete prior knowledge of the transition probability dynamics $P(s' | s, a)$ and reward function $R(s, a, s')$. The agent calculates expected outcomes across all possible futures prior to acting.

Model-Free Learning: The transition probabilities and reward mechanics are completely unknown. The agent cannot calculate expectations offline; it must directly interact with the physical environment, observe sample outcomes (s, a, r, s') , and iteratively update values based on observed returns.

- b. **Question:** In model-free reinforcement learning, an agent learns entirely from experience. Define the components of an experience tuple (s, a, r, s') .

Solution

- s : The current state of the agent before taking an action.
- a : The specific action selected and executed by the agent.
- r : The scalar numerical reward signal received from the environment.
- s' : The resulting next state reached by the agent after executing action a .

2. Step-by-Step Q-Learning Update

Scenario: Wingman UAV flaking maneuver with $Q(s, a) = 5.0$, $r = 2.0$, $\max_{a'} Q(s', a') = 8.0$, $\gamma = 0.5$, and $\alpha = 0.5$.

- a. **Question:** Calculate the **Target**: $\text{Target} = r + \gamma \max_{a'} Q(s', a')$.

Solution

$$\begin{aligned}\text{Target} &= r + \gamma \max_{a'} Q(s', a') \\ \text{Target} &= 2.0 + 0.5(8.0) = 2.0 + 4.0 = \mathbf{6.0}\end{aligned}$$

- b. **Question:** Calculate the **Temporal Difference (TD) Error**: $\delta = \text{Target} - Q(s, a)$. Interpret the physical meaning of the sign of δ .

Solution

$$\delta = \text{Target} - Q(s, a) = 6.0 - 5.0 = +1.0$$

Tactical Interpretation: Because $\delta > 0$, the operational outcome was **better than expected**. The immediate reward and estimated future potential together exceeded the agent's prior belief.

- c. **Question:** Compute the updated value $Q_{\text{new}}(s, a) = Q(s, a) + \alpha\delta$.

Solution

$$Q_{\text{new}}(s, a) = Q(s, a) + \alpha(\delta) = 5.0 + 0.5(1.0) = \mathbf{5.5}$$

3. Exploration vs. Exploitation Tradeoff (ϵ -Greedy)

- a. **Question:** Describe the mechanics of the ϵ -greedy action selection mechanism in terms of probabilities ϵ and $1 - \epsilon$.

Solution

- With probability $1 - \epsilon$, the agent **exploits** its learned knowledge by selecting the greedy action: $\arg \max_a Q(s, a)$.
- With probability ϵ , the agent **explores** by randomly selecting from all available alternative actions to discover new state transitions and verify if existing estimates are suboptimal.

- b. **Question:** The EW pod operator recommends setting $\epsilon = 0$ immediately upon deployment to maximize immediate jamming effectiveness. Explain the tactical danger of this greedy strategy.

Solution

Setting $\epsilon = 0$ forces 100% pure exploitation. Because initial Q-values are either initialized randomly or derived from limited reconnaissance, the early estimates are frequently inaccurate. With zero exploration, the system becomes permanently locked into suboptimal jamming frequencies, failing to discover superior channels that could successfully suppress adaptive radar threats.

Part 2: Student Independent Practice Solutions

A. Cyber-Defense Tactical Response

Scenario: Initial $Q(s, a) = 12.0$, $r = -4.0$, $\gamma = 0.8$, $\alpha = 0.25$.

State s' candidates: $Q(s', \text{Audit}) = 10.0$, $Q(s', \text{Restore}) = 6.0$, $Q(s', \text{BlockIP}) = 14.0$.

1. **Question:** Identify the value of $\max_{a'} Q(s', a')$ from the candidate actions in state s' .

Solution

$$\max_{a'} Q(s', a') = \max(10.0, 6.0, 14.0) = \mathbf{14.0} \quad (\text{associated with Action BlockIP})$$

2. **Question:** Compute the Q-learning Target value.

Solution

$$\text{Target} = r + \gamma \max_{a'} Q(s', a') = -4.0 + 0.8(14.0) = -4.0 + 11.2 = \mathbf{7.2}$$

3. **Question:** Compute the Temporal Difference error (δ). Did the outcome exceed or fall short of expectations?

Solution

$$\delta = \text{Target} - Q(s, a) = 7.2 - 12.0 = \mathbf{-4.8}$$

Because $\delta < 0$, the action outcome **fell short of expectations** (worse than anticipated due to the latency penalty).

4. **Question:** Calculate the updated value $Q(s, a)$.

Solution

$$Q(s, a) \leftarrow Q(s, a) + \alpha \delta = 12.0 + 0.25(-4.8) = 12.0 - 1.2 = \mathbf{10.8}$$

B. Hyperparameter Analysis: Learning Rate and TD Error

Scenario: Extreme anomaly event with $\delta = -20.0$.

1. **Question:** If the agent uses a conservative learning rate of $\alpha = 0.05$, what is the magnitude of the update adjustment?

Solution

$$\text{Adjustment} = \alpha \times \delta = 0.05 \times (-20.0) = \mathbf{-1.0}$$

The prior Q-value drops by only 1.0 unit, shielding the policy from overreacting to possible sensor noise.

- 2. Question:** If the learning rate were set to $\alpha = 0.90$, how would the agent react to this single observation? Contrast the tactical stability implications of both settings.

Solution

$$\text{Adjustment} = 0.90 \times (-20.0) = -18.0$$

Tactical Comparison:

- With $\alpha = 0.90$, the agent drastically overhauls its belief based on a single experience packet, risking policy instability and erratic oscillations if the event was an anomalous outlier.
- With $\alpha = 0.05$, learning is stable and robust against noise, but the agent adapts very slowly to genuine, structural changes in adversary tactics.

C. Full Update Formulation

Question: Write out the complete, single-line mathematical formula for Q-learning and label its four components.

Solution

$$Q(s, a) \leftarrow \underbrace{Q(s, a)}_{(1) \text{ Current belief}} + \underbrace{\alpha}_{(4) \text{ Learning rate}} \underbrace{\left[\overbrace{r + \gamma \max_{a'} Q(s', a')}^{(2) \text{ Target}} - Q(s, a) \right]}_{(3) \text{ TD Error } (\delta)}$$

1. **Current belief** ($Q(s, a)$): Prior estimate of long-term return.
2. **Target** ($r + \gamma \max_{a'} Q(s', a')$): Observed reward plus discounted future value.
3. **TD Error** (δ): Target minus current prediction.
4. **Adjustment** ($\alpha\delta$): Step-size shift applied to revise the old estimate.